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Influence of Alloying and Heat Treatment on the Corrosion Properties of Electrolytic Alloys on the Basis of Nickel

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Abstract. In this work the influence of molybdenum and heat treatment on the corrosion properties of nickel-based electrolytic alloys is considered. Structural features are studied and the interrelation between the structure and corrosion resistance of alloys is established.

INTRODUCTION

Corrosion control is one of the most significant issues in the oil and gas industry, as companies suffer serious losses from corrosion damage to oil and gas equipment. Protection against corrosion using electrolytic metals has been used for a long time, but the lifetime of electrolytic alloys has not been studied sufficiently. In matters of protection against corrosion, nickel-based electrolytic alloys are the most promising. In particular, alloys of the nickel-molybdenum system [1].

The paper considers electrolytic alloys with a 0–30 % content of molybdenum, obtained under different electrodeposition conditions, as well as those subjected to annealing.

Varying the parameters of electrodeposition, alloying and annealing, it is possible to obtain various structures of nickel-molybdenum alloys [2], and they can differ significantly from the state diagram characteristic of metallurgical alloys [3] due to significant differences in crystallization conditions.

The purpose of this work was to study the corrosion properties of coatings obtained under different conditions, degrees of alloying and heat treatment, and the relationship between corrosion resistance and the structure obtained.

SAMPLES AND EXPERIMENTAL TECHNIQUE

The following research methods were used during the work: production of coatings in factory and laboratory conditions, hardness testing on PMT-3M microhardness tester, diffraction methods were carried out using a DRON-7 X-ray diffractometer, structural studies were carried out with an Olympus GX51F optical microscope, electron microscopes JEOL JSM-6510A (raster microscopy) and Philips CM-12 (transmission microscopy), internal stresses were determined on a specially designed laboratory installation, corrosion tests were carried out gravimetrically by immersing the obtained samples in standard media, as well as in media consisting of bottom waters of the deposits of Western and Eastern Siberia.

RESULTS AND DISCUSSION

When analyzing the structure obtained in the course of electrodeposition [4], as well as its transformation during heat treatment, it was found that with an increase in the content of the alloying element-amorphizer in nickel-based electrolytic alloys, the following change in the phase state of the resulting coatings is observed: crystalline –

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amorphous-crystalline – amorphous. The deviation of electrodeposition conditions from thermodynamically equilibrium has the same effect on the structure of nickel-molybdenum alloys as the increase in the molybdenum concentration, namely, the degree of nonequilibrium of the formed alloy structure increases from structural to phase.

A feature of this work is that, when conducting corrosion tests, not only the standard media recommended by GOST were used, but also the produced water from the oil fields in Western and Eastern Siberia.

The composition of the media used in the tests is shown in Table 1.

TABLE 1. Chemical composition of oilfield corrosive media.

Medium (Field)	pH	Content of Ions, mg/l							Total Mineralization	
		Cl^-	HCO_3^-	SO_4^{2-}	Ca^{2+}	Mg^{2+}	Na^++K^+	Fe_{gen}	mg/l	mg-equiv/l
Kalchinskoye	8.03	5751	3416	5.6	72	7.8	4918.92	0.08	14171	419.1
Ety-Purovskoye	7	4468	549	8	276	12	2725	0	8038	258.8
Kuyumbinskoye	4.26	188150	28.06	170	42000	8640	57244.7	1.32	296234	10413.8

Tests of the obtained coatings were carried out immediately after electrodeposition and also after annealing at various temperatures.

When analyzing the phase composition of alloys obtained under different electrodeposition conditions [4], alloys obtained under "hard" conditions are of greatest interest, that is, alloys obtained at current densities close to the limiting ones (Table 2).

TABLE 2. Phase composition of nickel-molybdenum alloys after electrodeposition and annealing.

No.	Mo Content in the Alloys, wt. %	Phase Composition According to the Equilibrium Phase Diagram	Conditions of Electrodeposition			Conditions of Annealing of Alloys Electrodeposited atE>0.66 E_{lim}	
			E< 0.33E_{lim}	E=(0.33-0.66)E_{lim}	E > 0,66E_{lim}	(550°C – 10 h.)	(550°C –100h.)
1	5.0	Ni(Mo)	Ni(Mo)	Ni(Mo)	Ni(Mo)	Ni(Mo)	Ni(Mo)
2	12.5	Ni(Mo)	Ni(Mo)	Ni(Mo)	Ni(Mo)+$MoNi_4$	Ni(Mo)	Ni(Mo)
3	18.0	Ni(Mo)	Ni(Mo)	Ni(Mo)	Ni(Mo)+$MoNi_4$	Ni(Mo)	Ni(Mo)
4	25	Ni(Mo)+$MoNi_4$	Ni(Mo)+$MoNi_4$	Ni(Mo)*	Ni(Mo)**	Ni(Mo)*	Ni(Mo)+ $MoNi_4$
5	30	Ni(Mo)+$MoNi_4$	Ni(Mo)+ $MoNi_4$	Ni(Mo)**	Ni(Mo)**	Ni(Mo)*	Ni(Mo)+ $MoNi_4$

Ni(Mo)* supersaturated solid solutions of molybdenum in nickel;
Ni(Mo)** amorphous state of nickel-molybdenum alloys.

Figure 1 shows microstructures of pure nickel and alloys based on it directly obtained after electrodeposition.

Figure 2 presents the dependence of the change in corrosion rate on the heating temperature of Ni-Mo alloys with the content of the alloying element up to 5 %.

There are no significant changes in the rate of corrosion since during annealing stress is removed without transformation of the structure.

For alloys with an 18 % content of the alloying element (Fig. 3), a certain decrease in the corrosion rate is observed with an increase in the annealing temperature, which may be due to the transformation of the structure and dissolution of the intermetallide $MoNi_4$, whose reflections are not fixed by X-ray diffraction analysis of the annealed samples.

Alloys with a high molybdenum content (25–30 %) undergo significant structural changes during annealing. In these alloys, there is a transition from an amorphous state to a crystalline state, and thus active corrosion centers appear, which were not there immediately before electrodeposition. This causes a significant change in the corrosion rate of alloys after annealing (Fig. 4).

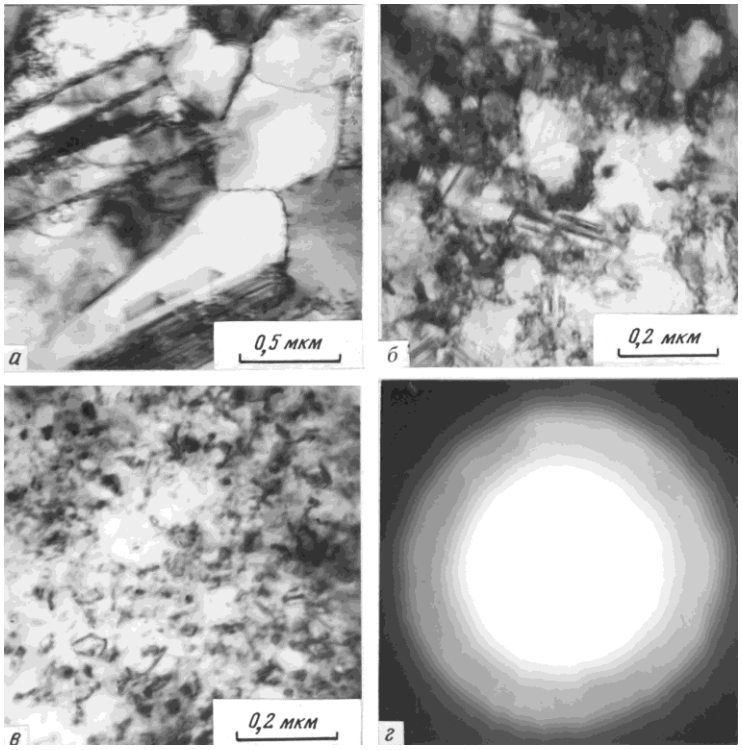

FIGURE 1. Microstructures of pure nickel (a), 12.5 % Ni-Mo alloys (b) and 25 % (c) and electron-diffraction pattern of Ni-25 % Mo alloy

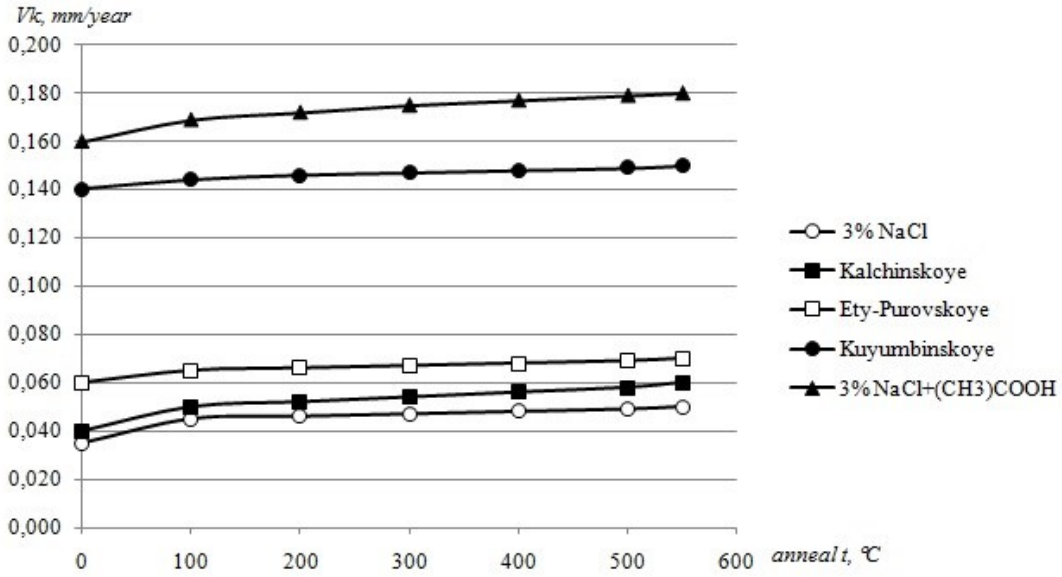

FIGURE 2. Dependence of corrosion rate on heating temperature for Ni-5 % Mo alloy

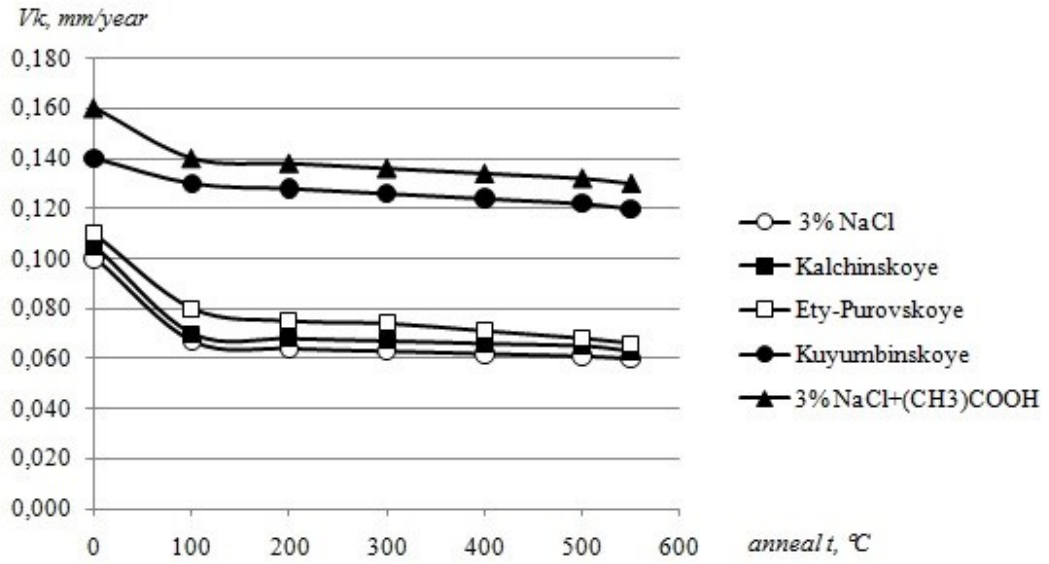

FIGURE 3. Dependence of corrosion rate on heating temperature for Ni-18 % Mo alloy

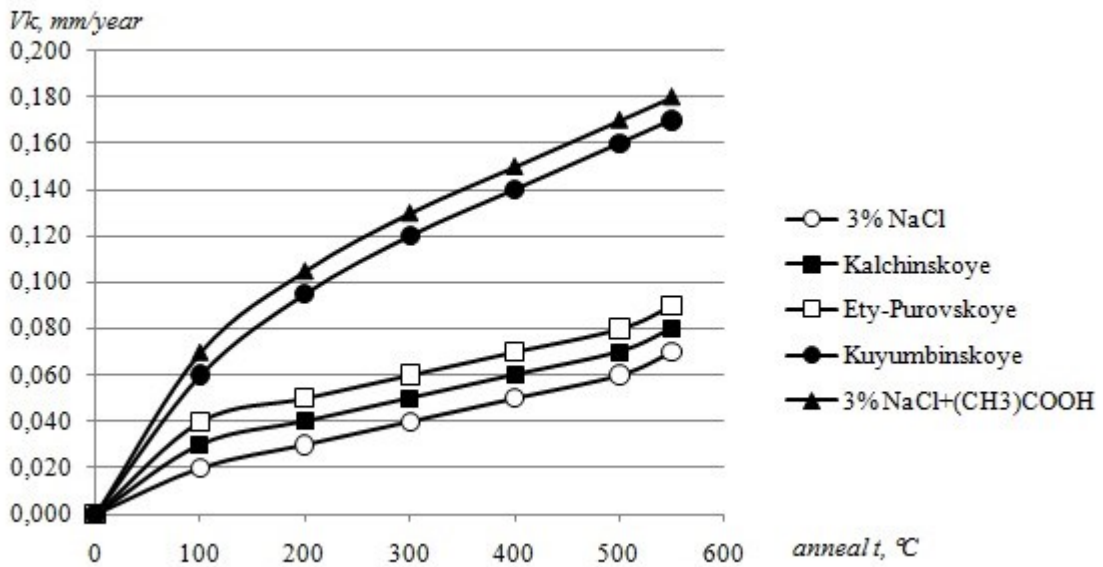

FIGURE 4. Dependence of corrosion rate on heating temperature for Ni-30%Mo alloy

CONCLUSIONS

The work shows change patterns of corrosion resistance of coatings depending on the content of the alloying element, taking into account the formed structure of coatings. Studies of the corrosion resistance of coatings are also given, depending on the conditions of heat treatment.

Using the obtained data, it is possible to select the optimal conditions of electrodeposition, alloying and heat treatment to ensure the best protection of oilfield equipment against corrosion.

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